SAMEEKSHA MAHESH

+1 (530)-760-0509 | sammahesh@ucdavis.edu | linkedin.com/in/sameeksham09 | github.com/sameeksham09

SUMMARY

Computer Science master's candidate at UC Davis with full-stack experience in Java, Python, Node.js and REST APIs. Developed and maintained microservices using Docker and AWS, reducing API response time by 20% and improving system reliability. Built OCR-based bank identification and end-to-end invoicing platforms under Agile methodologies. Strong collaborator with CI/CD and testing expertise.

EDUCATION

University of California, Davis — 2025 - 2027

Masters, Computer Science

- **GPA:** 3.7
- **Coursework:** Distributed Database Systems, Analysis of Software Artifacts

Dayanand Sagar College of Engineering — 2020 - 2024

Bachelors, Computer Science & Engineering

- **GPA:** 3.97
- **Coursework:** Data Structures and Algorithms, Operating System, Big Data, Cloud Computing, Object Oriented Programming

TECHNICAL SKILLS

- **Languages & Tools**: Java, Python, C++, SQL, JavaScript, Perl, Go
- **Frameworks & Tools**: Git, Spring Boot, Rest APIs, Elasticsearch, Microservices, Postman, Linux, React, NodeJS
- **Databases**: MySQL, PostgreSQL, MongoDB, Oracle
- **Cloud / DevOps**: AWS (EBS, EC2, Lambda), CI/CD, Docker, Kubernetes

EXPERIENCE

Perfios Software Solutions Pvt Ltd — Apr 2024 - Sep 2025

Software Development Engineer

- Redesigned **RESTful** microservices in Java and Spring Boot using object-oriented design principles, optimizing database queries and indexing to cut API response time by **20%** and improve system maintainability.
- Developed a **full-stack** invoicing platform with **Node JS** and React, implementing role-based access control (**RBAC**), conducting experiment design to validate reporting metrics, and partnering cross-functionally to accelerate data-driven decision making.
- Implemented a bank identification system with OCR for international banks, applying research engineering methods and writing unit tests to ensure robustness, boosting first-attempt accuracy by **35%.**
- Containerized and deployed services using Docker and established CI/CD pipelines via **GitHub** Actions and **Jenkins**, improving deployment reliability, reducing release cycles, and participating in code reviews to maintain quality.

Brane Enterprises Pvt. Ltd — Jun 2022 - Jan 2023

Automation Intern

- Automated **API workflows** using Postman and integrated them into the CI/CD pipeline, applying experiment design and data analysis to increase reliability, reduce manual testing effort by **15%**, and collaborating with cross-functional teams.
- Optimized client-facing automation tools in Node JS, working autonomously under Agile timelines and using **Git** for version control to improve platform performance and ensure thorough testing coverage.

PROJECTS

Intelligent Traffic Violation Detection System — Mar 2024 - Jun 2024

- Developed a real-time traffic violation detection system for helmet non-compliance, signal violations, and mobile phone usage using deep learning and **computer vision** (YOLOv3/v5/v8 + SORT).
- Integrated OCR (Tesseract) for automatic license plate recognition and implemented a **Streamlit** interface for video upload, violation visualization, and record storage.
- Achieved **73% mAP50** on validation datasets, enabling accurate and efficient monitoring of traffic violations.

Secure Healthcare Portal — Oct 2023 - Feb 2024

- Engineered a secure, **cloud-based** telemedicine platform for encrypted storage and sharing of medical records between patients and doctors, ensuring confidentiality, integrity, and reliable access.
- Implemented **RSA**-based encryption, key management, and two-factor authentication, and developed Flask REST APIs for secure data transactions, achieving 100% data confidentiality in tests.

Fake News Detection System — Feb 2023 - May 2023

- Developed a fake news detection system using **NLP** and ensemble **ML** models (Logistic Regression, SVM, Random Forest), achieving **96%** accuracy in classifying real vs. fake articles.
- Engineered a text preprocessing pipeline (tokenization, stopword removal, TF-IDF vectorization) and fine-tuned model hyperparameters to maximize prediction precision and robustness.
- Designed and deployed an interactive **web interface** for real-time article submission and credibility scoring, enhancing user transparency and engagement.